Lesson 2.4 Confounding in the 3^k Factorial Design

Introduction

Even when a single replicate of the 3^k design is considered, the design requires so many runs that it is unlikely that all 3^k runs can be made under uniform conditions. Thus, confounding in blocks is often necessary. The 3^k design may be confounded in 3^p incomplete blocks, where $p < k$. Thus, these designs may be confounded in three blocks, nine blocks, and so on.

The 3^k Factorial Design in Three Blocks

Suppose that we wish to confound the 3^k design in three incomplete blocks. These three blocks have two degrees of freedom among them; thus, there must be two degrees of freedom confounded with blocks. Recall that in the 3^k factorial series each main effect has two degrees of freedom. Furthermore, every two-factor interaction has four degrees of freedom and can be decomposed into two components of interaction (e.g., AB and AB^2), each with two degrees of freedom; every three-factor interaction has eight degrees of freedom and can be decomposed into four components of interaction (e.g., ABC, ABC^2, AB^2C, and AB^2C^2), each with two degrees of freedom; and so on. Therefore, it is convenient to confound a component of interaction with blocks.

The general procedure is to construct a defining contrast

$$L = \alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_k x_k$$

where α_i represents the exponent on the ith factor in the effect to be confounded and x_i is the level of the ith factor in a particular treatment combination. For the 3^k series, we have $\alpha_i = 0$, 1, or 2 with the first nonzero α_i being unity, and $x_i = 0$ (low level), 1 (intermediate level), or 2 (high level). The treatment combinations in the 3^k design are assigned to blocks based on the value of $L(mod3)$. Because $L(mod3)$ can take on only the values 0, 1, or 2, three blocks are uniquely defined. The treatment combinations satisfying $L = 0(mod3)$ constitute the principal block. This block will always contain the treatment combination $00 \cdots 0$.

For example, suppose we wish to construct a 3^2 factorial design in three blocks. Either component of the AB interaction, AB or AB^2, may be confounded with blocks. Arbitrarily choosing AB^2, we obtain the defining contrast

$$L = x_1 + 2x_2$$

The value of L (mod 3) of each treatment combination may be found as follows:

$$
\begin{aligned}
00 &: L = 1(0) + 2(0) = 0(mod3) = 0 \\
01 &: L = 1(0) + 2(1) = 2(mod3) = 2 \\
02 &: L = 1(0) + 2(2) = 4(mod3) = 1 \\
10 &: L = 1(1) + 2(0) = 1(mod3) = 1 \\
11 &: L = 1(1) + 2(1) = 3(mod3) = 0 \\
12 &: L = 1(1) + 2(2) = 5(mod3) = 2 \\
20 &: L = 1(2) + 2(0) = 2(mod3) = 2 \\
21 &: L = 1(2) + 2(1) = 4(mod3) = 1 \\
22 &: L = 1(2) + 2(2) = 6(mod3) = 0
\end{aligned}
$$

The blocks are shown below.

Block 1	Block 2	Block 3
00	10	01
11	21	12
22	02	20

The elements in the principal block form a group with respect to addition modulus 3. Referring to the figure, we see that $11 + 11 = 22$ and $11 + 22 = 00$. Treatment combinations in the other two blocks may be generated by adding, modulus 3, any element in the new block to the elements of the principal block. Thus, we use 10 for block 2 and obtain

An example: 3^2 design in 3 blocks

We illustrate the statistical analysis of the 3^2 design confounded in three blocks by using the following data, which come from the single replicate of the 3^2 design.

	Block 1	Block 2	Block 3
	00 = 4	10 = −2	01 = 5
	11 = −4	21 = 1	12 = −5
	22 = 0	02 = 8	20 = 0
Block totals =	0	7	0

Using conventional methods for the analysis of factorials we get the following sums of squares.

```r
library(tidyverse)
library(kableExtra)
Block <- c(rep(1,3), rep(2,3),rep(3,3))
A <- c( 0, 1, 2, 1, 2, 0, 0, 1, 2)
B <- c(0, 1, 2, 0, 1, 2, 1, 2, 0)
Y <- c( 4, -4, 0, -2, 1, 8, 5, -5, 0)
df <- data.frame(cbind(Block, A, B, Y))
df <- df %>%
  mutate(Block = factor(Block),
         A = factor(A),
         B = factor(B))
out1 <- aov(Y ~ Block + A*B, data = df)
kable(anova(out1), digits = 4)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Block	2	10.8889	5.4444	NaN	NaN
A	2	131.5556	65.7778	NaN	NaN
B	2	0.2222	0.1111	NaN	NaN
A:B	2	2.8889	1.4444	NaN	NaN
Residuals	0	0.0000	NaN	NA	NA

However, SS_{Blocks} is exactly equal to the AB^2 component of interaction. To see this, write the observations as follows:

		Factor *B*		
		0	**1**	**2**
	0	4	5	8
Factor *A*	1	−2	−4	−5
	2	0	1	0

Recall from the previous lesson that the $I(AB)$ or AB^2 component of the AB interaction may be found by computing the sum of squares between the left-to-right diagonal totals in the above layout. This yields

$$SS_{AB^2} = \frac{0^2 + 0^2 + 7^2}{3} - \frac{7^2}{9} \approx 10.889$$

which is identical to SS_{Blocks}.

We now look at a slightly more complicated design—a 3^3 factorial confounded in three blocks of nine runs each. The AB^2C^2 component of the three-factor interaction will be confounded with blocks. The defining contrast is

$$L = x_1 + 2x_2 + 2x_3$$

It is easy to verify that the treatment combinations 000, 012, and 101 belong in the principal block. The remaining runs in the principal block are generated as follows:

$$
\begin{aligned}
012 + 012 &= 021 \\
101 + 101 &= 202 \\
101 + 012 &= 110 \\
101 + 021 &= 122 \\
012 + 202 &= 211 \\
021 + 202 &= 220
\end{aligned}
$$

To find the runs in another block, note that the treatment combination 200 is not in the principal block. Thus, the elements of block 2 are

$$
\begin{aligned}
200 + 000 &= 200 \\
200 + 102 &= 212 \\
200 + 101 &= 001 \\
200 + 021 &= 221 \\
200 + 202 &= 102 \\
200 + 110 &= 010 \\
200 + 122 &= 022 \\
200 + 211 &= 111 \\
200 + 220 &= 120
\end{aligned}
$$

The blocks are shown in the figure below. Verify the runs in Block 3.

Block 1	Block 2	Block 3
000	200	100
012	212	112
101	001	201
202	102	002
021	221	121
110	010	210
122	022	222
211	111	011
220	120	020

The analysis of variance for this design is shown below.

Analysis of Variance for a 3^3 Design with AB^2C^2 Confounded

Source of Variation	Degrees of Freedom
Blocks (AB^2C^2)	2
A	2
B	2
C	2
AB	4
AC	4
BC	4
Error ($ABC + AB^2C + ABC^2$)	6
Total	26

Through the use of this confounding scheme, information on all the main effects and two-factor interactions is available. The remaining components of the three-factor interaction (ABC, AB^2C, and ABC^2) are combined as an estimate of error. The sum of squares for those three components could be obtained by subtraction. In general, for the 3^k design in three blocks, we would always select a component of the highest order interaction to confound with blocks. The remaining unconfounded components of this interaction could be obtained by computing the k-factor interaction in the usual way and subtracting from this quantity the sum of squares for blocks.

Another example

An experiment was performed to study the effect of three different types of 32-ounce bottles (A) and three different shelf types (B)—smooth permanent shelves, end-aisle displays with grilled shelves, and beverage coolers—on the time it takes to stock ten 12-bottle cases on the shelves. Three workers (factor C) were employed in the experiment, and two replicates of a 3^3 factorial design were run. The observed time data are shownin the following table.

		Replicate I					Replicate II		
Worker	Bottle Type	Per-manent	End Aisle	Cooler			Per-manent	End Aisle	Cooler
	Plastic	3.45	4.14	5.80		Plastic	3.36	4.19	5.23
1	28-mm glass	4.07	4.38	5.48	1	28-mm glass	3.52	4.26	4.85
	38-mm glass	4.20	4.26	5.67		38-mm glass	3.68	4.37	5.58
	Plastic	4.80	5.22	6.21		Plastic	4.40	4.70	5.88
2	28-mm glass	4.52	5.15	6.25	2	28-mm glass	4.44	4.65	6.20
	38-mm glass	4.96	5.17	6.03		38-mm glass	4.39	4.75	6.38
	Plastic	4.08	3.94	5.14		Plastic	3.65	4.08	4.49
3	28-mm glass	4.30	4.53	4.99	3	28-mm glass	4.04	4.08	4.59
	38-mm glass	4.17	4.86	4.85		38-mm glass	3.88	4.48	4.90

1. Analyze the data using standard methods for factorial experiments and draw conclusions.

2. Set up a design for conducting the experiment over three blocks with AB^2C confounded with blocks. Analyze the data.

The 3^k Factorial Design in Nine Blocks

In some experimental situations, it may be necessary to confound the 3k design in nine blocks. Thus, eight degrees of freedom will be confounded with blocks. To construct these designs, we choose two components of interaction, and, as a result, two more will be confounded automatically, yielding the required eight degrees of freedom. These two are the generalized interactions of the two effects originally chosen. In the 3^k system, the generalized interactions of two effects (e.g., P and Q) are defined as PQ and PQ^2 (or P^2Q).

The two components of interaction initially chosen yield two defining contrasts

$$L_1 = \alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_k x_k = u(mod3), \quad u = 0, 1, 2$$
$$L_2 = \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k = h(mod3), \quad h = 0, 1, 2$$

where α_i and β_j are the exponents in the first and second generalized interactions, respectively, with the convention that the first nonzero α_i and β_j are unity. The defining contrasts L_1 and L_2 imply nine simultaneous equations specified by the pair of values for L_1 and L_2. Treatment combinations having the same pair of values for (L_1, L_2) are assigned to the same block.

The principal block consists of treatment combinations satisfying $L_1 = L_2 = 0(mod3)$. The elements of this block form a group with respect to addition modulus 3; thus, the scheme given the previous lessons can be used to generate the blocks. As an example, consider the 3^4 factorial design confounded in nine blocks of nine runs each. Suppose we choose to confound ABC and AB^2D^2. Their generalized interactions

$$(ABC)(AB^2D^2) = A^2B^3CD^2 = (A^2B^3CD^2)^2 = AC^2D$$
$$(ABC)(AB^2D^2)^2 = A^3B^5CD^4 = B^2CD = (B^2CD)^2 = BC^2D^2$$

are also confounded with blocks. The defining contrasts for ABC and AB^2D^2 are

$$L_1 = x_1 + x_2 + x_3$$
$$L_2 = x_1 + 2x_2 + 2x_4$$

The nine blocks may be constructed by using these defining contrasts and the group-theoretic property of the principal block. The design is shown below.

Block 1	Block 2	Block 3	Block 4	Block 5	Block 6	Block 7	Block 8	Block 9
0000	0001	2000	0200	0020	0010	1000	0100	0002
0122	0120	2122	0022	0112	0102	1122	0222	0121
0211	0212	2211	0111	0201	0221	1211	0011	0210
1021	1022	0021	1221	1011	1001	2021	1121	1020
1110	1111	0110	1010	1100	1120	2110	1210	1112
1202	1200	0202	1102	1222	1212	2202	1002	1201
2012	2010	1012	2212	2002	2022	0012	2112	2011
2101	2102	1101	2001	2121	2111	0101	2201	2100
2220	2221	1220	2120	2210	2200	0220	2020	2222
$(L_1, L_2) = (0,0)$	(0,2)	(2,2)	(2,1)	(2,0)	(1,6)	(1,1)	(1,2)	(0,1)

The 3^k Factorial Design in 3^p Blocks

The 3^k factorial design may be confounded in 3^p blocks of 3^{k-p} observations each, where $p < k$. The procedure is to select p independent effects to be confounded with blocks. As a result, exactly $(3^p - 2p - 1)/2$ other effects are automatically confounded. These effects are the generalized interactions of those effects originally chosen. As an illustration, consider a 3^7 design to be confounded in 27 blocks. Because $p = 3$, we would select three independent components of interaction and automatically confound others. Suppose we choose ABC^2DG, BCE^2F^2G, and $BDEFG$. Three defining contrasts can be constructed from these effects, and the 27 blocks can be generated by the methods previously described. The other 10 effects confounded with blocks are

$$
\begin{aligned}
(ABC^2DG)(BCE^2F^2G) &= AB^2DE^2F^2G^2 \\
(ABC^2DG)(BCE^2F^2G)^2 &= AB^3C^4DE^4F^4G^3 = ACDEF \\
(ABC^2DG)(BDEFG) &= AB^2C^2D^2EFG^2 \\
(ABC^2DG)(BDEFG)^2 &= AB^3C^2D^3E^2F^2G^3 = AC^2E^2F^2 \\
(BCE^2F^2G)(BDEFG) &= B^2CDE^3F^3G^2 = BC^2D^2G \\
(BCE^2F^2G)(BDEFG)^2 &= B^3CD^2E^4F^4G^3 = CD^2EF \\
(ABC^2DG)(BCE^2F^2G)(BDEFG) &= AB^3C^3D^2E^3F^3G^3 = AD^2 \\
(ABC^2DG)^2(BCE^2F^3G)(BDEFG) &= A^2B^4C^5D^3G^4 = AB^2CG^2 \\
(ABC^2DG)(BCE^2F^2G)^2(BDEFG) &= ABCD^2E^2F^2G \\
(ABC^2DG)(BCE^2F^2G)(BDEFG)^2 &= ABC^3D^3E^4F^4G^4 = ABEFG
\end{aligned}
$$

This is a huge design requiring $3^7 = 2187$ observations arranged in 27 blocks of 81 observations each.